

Operating Instructions

Step 1: System Power-Up & Initialization

Connect the Arduino microcontroller to a stable power source via the USB interface.

The system will automatically initialize all integrated peripheral hardware devices.

Step 2: Real-Time Environmental Automation (Background Process)

The system continuously samples and monitors desert ambient conditions in the background during operation:

Night Migration Lighting: If the LDR sensor detects ambient darkness, the system will instantly trigger the Yellow LED to simulate guiding torches. The LED will turn off automatically when daylight is restored.

Temperature Sensor: The temperature sensor on continuously monitors ambient heat. If the temperature registers a value above 30°C, the Temperature Red LED will illuminate.

Step 3: Weight Verification Protocol (Weighing Station)

Every traveler must step onto the digital scale platform to weigh their belongings before crossing.

Conditional Logic Execution:

Pass (< 5 kg): If the measured weight is below 5 kg, the Weight Green LED will light up.

Fail (> 5 kg): If the gear exceeds the 5 kg threshold, the Weight Red LED will illuminate. er to reduce weight, halting further progression until the load is acceptable.

Step 4: Identity Validation & Gate Activation (Crossing the Red Sea)

Upon passing the weight screening, a verification question will appear on the LCD screen.

The user must point the IR Remote Control at the IR Receiver module and input the correct numerical answer.

Validation Outcomes:

Correct Answer: The LCD will display a success message, and the Servo Motor will rotate to open the passage gate (parting the Red Sea). The gate remains open for a set window before automatically returning to its secure closed position.

Incorrect Answer: The gate remains firmly locked, and an error message is prompted on the screen.

Step 5: Sequential Migration Closure & System Termination

After the entire caravan has safely crossed the checkpoint, the final operator (Moses or an authorized leader) must engage the physical Termination Push Button.

Critical Safety & Operational Constraints

Sensor Hygiene: Ensure the surface of the Load Cell platform is completely clear of external objects before turning the system on to allow for proper initial software calibration.

Environmental Stability: For reliable LDR and Temperature monitoring, avoid placing the system under erratic or flickering artificial light sources.

Mechanical Safety: Do not apply external physical force or hang excessive weight on the fragile Servo gate mechanism to avoid stripped internal gears.